2.2 Parallel-Plate Capacitor: Laplace's Equation

Part IB Electromagnetism provides background but is not necessary: the project is self-contained.

1 Introduction

According to elementary electromagnetic theory, an isolated parallel-plate capacitor [without dielectric spacer] has plate charge q proportional to potential difference V, the constant of proportionality –'capacitance' – being given by $q/V = \epsilon_0 A/d$ where A is the area of the plates, d their separation, and ϵ_0 the 'permittivity of free space' (a constant). This assertion is usually justified by arguing that the electric field $\mathbf{E}$ is approximately uniform between the conducting plates and much weaker elsewhere, except near the plate edges whose effects may be ignored.

In this project it is assumed that the plates are rectangles of length l_z, width l_x and negligible thickness, occupying $-l_x/2 < x < l_x/2$, $y = \pm d/2$, $-l_z/2 < z < l_z/2$. If in addition $l_z \gg l_x$, then away from the ends $z = \pm l_z/2$ the electric field should be approximately two-dimensional: $\mathbf{E} \approx -\nabla\phi(x, y)$ where the potential ϕ satisfies the two-dimensional Laplace equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0 \tag{1}$$

with boundary conditions

$$\begin{aligned} \phi(x,y) &= +V/2 \quad \text{on the top plate: } y = d/2,\ -l_x/2 \leqslant x \leqslant l_x/2 \\ \phi(x,y) &= -V/2 \quad \text{on the bottom plate: } y = -d/2,\ -l_x/2 \leqslant x \leqslant l_x/2 \end{aligned} \tag{2}$$

and

$$\phi(x, y) \to 0 \text{ as } |x| \to \infty \text{ and/or } |y| \to \infty \tag{3}$$

Then from Gauss's Law, the charge per unit length in z on the top (+) or bottom (−) plate is

$$\hat{q}_\pm \equiv \epsilon_0 \oint_{\gamma_\pm} \mathbf{E}\,.\,\mathbf{n}\,\mathrm{d}s = \epsilon_0 \oint_{\gamma_\pm} -\nabla\phi\,.\,(\mathrm{d}y, -\mathrm{d}x) = -\epsilon_0 \oint_{\gamma_\pm} \frac{\partial\phi}{\partial x}\,\mathrm{d}y - \frac{\partial\phi}{\partial y}\,\mathrm{d}x \tag{4}$$

where γ_+ (γ_-) is any simple closed contour in the x-y-plane encircling the top (bottom) plate (and not the other); note that $\phi(x, y)$ will be [even in x and] odd in y, implying that $\hat{q}_- = -\hat{q}_+$. Solutions of this two-dimensional 'strip-capacitor' problem give some idea of the validity of the approximation that the electric field is uniform between the plates [$\phi \approx Vy/d$, $\mathbf{E} \approx (0, -V/d, 0)$] and ignorable elsewhere, which leads to

$$\hat{q}_\pm \approx \pm\frac{\epsilon_0\, l_x}{d}\, V\ . \tag{5}$$

In terms of dimensionless variables defined by

$$x = \tfrac{1}{2}dX\ , \quad y = \tfrac{1}{2}dY\ , \quad \phi(x, y) = V\Phi(X, Y)\ , \quad \hat{q}_\pm = \pm\epsilon_0 V Q_\infty \tag{6}$$

the problem becomes

$$\frac{\partial^2\Phi}{\partial X^2} + \frac{\partial^2\Phi}{\partial Y^2} = 0\ , \tag{7}$$

$$\Phi(X, 1) = +\tfrac{1}{2}\ , \quad \Phi(X, -1) = -\tfrac{1}{2} \quad \text{for } -L \leqslant X \leqslant L, \tag{8}$$

$$\Phi(X, Y) \to 0 \text{ as } |X| \to \infty \text{ and/or } |Y| \to \infty \tag{9}$$

where $L = l_x/d$ is the width-to-separation ratio of the plates. The non-dimensionalised charge per unit length on the top plate is given by

$$Q_\infty = -\oint_{\Gamma_+} \frac{\partial\Phi}{\partial X}\,\mathrm{d}Y - \frac{\partial\Phi}{\partial Y}\,\mathrm{d}X \tag{10}$$

for any simple closed contour Γ_+ in the X-Y-plane encircling the top plate (and not the other).

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2 Analytic Solution

The problem (7)–(9) can be solved analytically by conformal mapping, but only in awkward implicit form involving elliptic integrals: see for example reference [1]*. However, if L is 'large' it may be expected that the 'fringing field' near the plate ends, say those at $X = +L$, can be approximated by that for semi-infinite plates, with (8) replaced by

$$\Phi(X,1) = +\tfrac{1}{2}\ , \quad \Phi(X,-1) = -\tfrac{1}{2} \quad \text{for } -\infty \leqslant X \leqslant L. \tag{11}$$

For this, conformal mapping yields the simple, albeit implicit, analytic solution

$$Z \equiv X + \mathrm{i}Y = L - 2\mathrm{i}W + \pi^{-1}\left(1 + \mathrm{e}^{-2\pi\mathrm{i}W}\right)\ , \quad W = -\Phi + \mathrm{i}\Psi \tag{12}$$

where $\Psi(X,Y)$ is a 'harmonic conjugate' of $(-\Phi)$ satisfying the Cauchy-Riemann equations $(-\Phi)_X = \Psi_Y$, $(-\Phi)_Y = -\Psi_X$; physically it is a 'flux function' †, lines of constant Ψ being everywhere tangent to the (non-dimensionalised) electric field $\nabla(-\Phi) \equiv \nabla \times (0,0,\Psi)$.

Question 1 For the semi-infinite-plate solution (12), plot some equipotentials [contours of constant $\Phi \in [-1/2, 1/2]$, parametrised by $\Psi \in (-\infty, \infty)$] and electric field lines [contours of constant Ψ, parametrised by Φ] in the $(X–L, Y)$-plane.

By computing

$$-\frac{\partial\Phi}{\partial X} + \mathrm{i}\frac{\partial\Phi}{\partial Y} = \frac{\mathrm{d}W}{\mathrm{d}Z} = \frac{1}{\mathrm{d}Z/\mathrm{d}W} \tag{13}$$

as a function of W, and then setting $\Phi = \frac{1}{2}$ and approximating (12) and (13) for small Ψ and for large positive Ψ, show that on the upper surface of the top plate (where $\Phi = \frac{1}{2}$, $\infty > \Psi > 0$), the Y-component – the only non-zero component – of (non-dimensionalised) electric field has the asymptotic behaviours

$$-\left.\frac{\partial\Phi}{\partial Y}\right|_{Y=1^+} \sim \begin{cases} a(L-X)^{-1/2} & \text{as } X \to L \ \ [\Psi \to 0] \\ b(L-X)^{-1} & \text{as } X \to -\infty \ \ [\Psi \to \infty] \end{cases} \tag{14}$$

where a and b are constants to be determined. How does it behave in these limits on the lower surface of the top plate (where $\Phi = \frac{1}{2}$, $-\infty < \Psi < 0$)?

3 Numerical Method

The problem (7)–(9) is to be restricted to a finite rectangular domain $\mathcal{D} = \{|X| < D_X, |Y| < D_Y\}$, with the final boundary condition (9) replaced by $\Phi(X,Y) = 0$ on the domain boundaries $X = \pm D_X$ and $Y = \pm D_Y$ (which physically might represent an earthed box enclosing the capacitor), and solved numerically using the following *finite-difference method*.

Consider the mesh of grid points $(X,Y) = (mh, nh)$ for integers $m \in [-N_X, N_X]$ and $n \in [-N_Y, N_Y]$, where the mesh spacing $h = 1/N_h$ for some integer N_h, and $D_X = hN_X$, $D_Y = hN_Y$. The discretised approximation to $\Phi(mh, nh)$ will be denoted by $\Phi_{m,n}$. [If using MATLAB, this might be stored in the (N_X+m+1, N_Y+n+1) element of an array, or in the $(m+1, n)$ element if solving only in the first quadrant – see below.]

*This reference uses 'capacitance' to mean what in the project is capacitance *per unit length* (i.e. the ratio of charge *per unit length* to potential difference), and denotes by R and C what in the project are L and $Q_\infty/4\pi$.

†It is the analogue of a 'streamfunction' in fluid dynamics, where a two-dimensional incompressible velocity field is customarily represented as $\nabla \times (0,0,\Psi)$, and if irrotational as $\nabla\Phi$ where Φ is a 'velocity potential' – note the different sign convention. [The solution (12) is sometimes credited to Maxwell, but as he himself acknowledges in reference [2], it had been found previously by Helmholtz for a mathematically equivalent fluid-dynamic problem (the 'Borda mouthpiece').]

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Question 2 Show that for 'small' h

$$\frac{1}{h^2}\left[\Phi_{i-1,j}+\Phi_{i+1,j}+\Phi_{i,j-1}+\Phi_{i,j+1}-4\Phi_{i,j}\right] \;=\; 0 \tag{15}$$

is a discretised approximation to the Laplace equation (7).

Equation (15), together with the boundary conditions

$$\Phi_{i,N_h}=\tfrac{1}{2}\ , \quad \Phi_{i,-N_h}=-\tfrac{1}{2} \quad \text{when } -L/h\leqslant i\leqslant L/h \tag{16}$$

and

$$\Phi_{i,j}=0 \quad \text{when } i=\pm N_X \text{ and/or } j=\pm N_Y, \tag{17}$$

can be solved using an *iteration scheme*: an initial guess $\Phi_{i,j}^{(0)}$ is made for $\Phi_{i,j}$ (consistent with the boundary conditions, of course) and a sequence of approximations $\left\{\Phi_{i,j}^{(k)}\ ,\ k\geqslant 0\right\}$ is computed iteratively, the $\Phi_{i,j}^{(k+1)}$ being determined (collectively) from the $\Phi_{i,j}^{(k)}$. One such scheme, known as *Jacobi iteration*, is

$$\Phi_{i,j}^{(k+1)} \;=\; \frac{1}{4}\left[\Phi_{i-1,j}^{(k)}+\Phi_{i+1,j}^{(k)}+\Phi_{i,j-1}^{(k)}+\Phi_{i,j+1}^{(k)}\right] \tag{18}$$

A better method is *Gauss-Seidel iteration* which replaces old values with new as soon as they become available. Assuming that $\Phi_{i,j}$ is updated in order of increasing i and j, the scheme becomes

$$\Phi_{i,j}^{(k+1)} \;=\; \frac{1}{4}\left[\Phi_{i-1,j}^{(k+1)}+\Phi_{i+1,j}^{(k)}+\Phi_{i,j-1}^{(k+1)}+\Phi_{i,j+1}^{(k)}\right]\ . \tag{19}$$

Further improvement may be obtained via *Successive Over-relaxation* (SOR),

$$\Phi_{i,j}^{(k+1)}=(1-\omega)\Phi_{i,j}^{(k)} \;+\; \frac{\omega}{4}\left[\Phi_{i-1,j}^{(k+1)}+\Phi_{i+1,j}^{(k)}+\Phi_{i,j-1}^{(k+1)}+\Phi_{i,j+1}^{(k)}\right]\ . \tag{20}$$

where the 'relaxation parameter' ω must be less than 2 for stability. A well-chosen $\omega\in(1,2)$ (*over*-relaxation, as opposed to *under*-relaxation with $0<\omega<1$) may give significantly faster convergence than (19). More details can be found in introductory numerical-methods texts, such as references [3]–[6].

In fact, since for this configuration $\Phi(X,Y)$ is even in X and odd in Y, it is only necessary to solve (20) in the first quadrant, i.e. for $0\leqslant i<N_X$ and $0<j<N_Y$, with $\Phi_{i,0}=0$ and $\Phi_{-1,j}$ equated to $\Phi_{1,j}$.

Programming Task: Write a program to implement the SOR method (20). As a criterion for when to stop iterations, define the residual

$$r_k=\frac{1}{N_{pts}}\sum_i\sum_j\left|\Phi_{i,j}^{(k)}-\Phi_{i,j}^{(k-1)}\right| \tag{21}$$

where the sum is over all interior points, and N_{pts} is the number of them. The k-th iteration $\Phi^{(k)}$ is to be deemed sufficiently accurate if r_k is less than a suitable tolerance of your choice, e.g. $10^{-2}, 10^{-4}, \ldots$.

Before beginning coding, please read below to have in mind what else your program(s) will be expected to do.

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Question 3 Run your program with $\omega = 1$, i.e. using the basic Gauss-Seidel scheme (19), and each of the following five sets of inputs:

set	L	D_x	D_y	h
A	1	2	2	0.5
B	2	4	4	0.5
C	2	4	4	0.25
D	2	4	4	0.125
E	2	8	8	0.5

First confirm that for set A, correct to three decimal places,

$$\Phi_{0,1} = \Phi_{0,3} = 0.244 \ , \quad \Phi_{1,1} = \Phi_{1,3} = 0.238 \ , \quad \Phi_{2,1} = \Phi_{2,3} = 0.208 \ , \tag{22}$$

$$\Phi_{3,1} = \Phi_{3,3} = 0.095 \ , \quad \Phi_{3,2} = 0.173 \ . \tag{23}$$

Then for each of the sets B, C, D and E, (i) state the values of $\Phi_{i,j}$ at $(X, Y) = (0, 2)$ and at $(X, Y) = (2, 2)$ correct to at least three decimal places, and (ii) plot

- the approximation $\Phi_{0,j}$, $0 \leqslant j \leqslant N_Y$ to $\Phi(0, Y = jh)$ against Y $(\geqslant 0)$,
- the approximation to $\Phi(2, Y)$ - against Y $(\geqslant 0)$,
- an approximation to the Y-component – the only non-zero component – of electric field $-\partial\Phi/\partial Y$ against X $(\geqslant 0)$ on the midplane $Y = 0$ and on both upper and lower surfaces of the top plate at $Y = 1$, e.g. using

$$\frac{\partial \phi}{\partial Y}(X, Y) \approx \frac{\Phi(X, Y + h) - \Phi(X, Y - h)}{2h} \tag{24}$$

or

$$\frac{\partial \phi}{\partial Y}(X, Y) \approx \frac{-\Phi(X, Y + 2h) + 4\Phi(X, Y + h) - 3\Phi(X, Y)}{2h} \ . \tag{25}$$

The corresponding plots for sets B, C and D, and those for sets B and E, should be superposed to display the variation with grid spacing and with domain size. How have you ensured the required three-decimal-place accuracy?

Question 4 Try different values of $\omega \in (1, 2)$. Investigate how the optimal ω, for which convergence to the stopping condition takes the fewest iterations, varies with the size and shape of the grid. (It may be instructive to carry out computations with other values of h.) Does it also depend on the tolerance, and/or the initial guess $\Phi^{(0)}_{i,j}$?

Question 5 The (non-dimensionalised) charge on the top plate for the capacitor in the finite (earthed-box) domain $\mathcal{D}$ is still given by the integral (10), now denoted by $Q_{\mathcal{D}}$. Explain carefully why this integral can be discretised as

$$\begin{aligned} Q_{\mathcal{D}} \approx & \sum_{i=M_L}^{M_R} [\Phi_{i,N_B} - \Phi_{i,N_B-1} + \Phi_{i,N_T} - \Phi_{i,N_T+1}] \\ & + \sum_{j=N_B}^{N_T} [\Phi_{M_L,j} - \Phi_{M_L-1,j} + \Phi_{M_R,j} - \Phi_{M_R+1,j}] \end{aligned} \tag{26}$$

for appropriate values of M_L, M_R, N_B and N_T.

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Programming Task: Write code to compute the approximation (26) to $Q_{\mathcal{D}}$, given a solution $\Phi_{i,j}$ from your SOR program. Use your judgement about whether to have a separate program which does this or just add this functionality to code already written. It may be helpful to maintain some flexibility so that you can also compute the charge elsewhere.

Question 6 Compute such a numerical approximation to $Q_{\mathcal{D}}$ for sets B, C and D in Question 3, stating clearly the values of M_L, M_R, N_B and N_T used. What do these computations indicate about the 'discretisation errors', i.e. the difference between what is obtained using a discrete lattice of points and what would be measured in the continuum in a laboratory experiment? (For example, can you identify the order of accuracy as $h \to 0$?) You may wish to try smaller values of h, and/or extrapolation to $h = 0$. Mention any checks done to confirm the correctness of the computations.

Question 7 What is the effect of using the boundary condition $\Phi = 0$ on the sides of the finite domain compared to the original boundary condition that Φ vanishes at infinity? To investigate this, observe what happens as the boundaries at $X = \pm D_X$ and/or those at $Y = \pm D_Y$ are moved closer to the plates. It would be appropriate to include a few more plots to illustrate your answer.

Question 8 Investigate the dependence of Q_∞, the infinite-domain (top-)plate charge per unit length non-dimensionalised by $\epsilon_0 V$ (or the capacitance per unit length non-dimensionalised by ϵ_0), on the plate-width-to-separation ratio L, displaying results in table and/or graphical form. You should do this by computing, for various values of L, a discretised approximation to the solution on a finite domain with (possibly L-dependent) values of D_X and D_Y chosen to give, subject to reasonable computing time, as good an approximation as possible to an infinite domain, while still having h sufficiently small for acceptable accuracy – or maybe use more than one value of h and extrapolate to $h = 0$. Assess the extent to which this has been achieved, e.g. by observing the effect of varying these parameters.[‡] You might also note that the infinite-domain solution can be shown to have far-field 'dipole' behaviour

$$\Phi \sim \frac{Q_\infty}{\pi} \frac{Y}{X^2 + Y^2} \quad \text{as } |X| \to \infty \text{ and/or } |Y| \to \infty \ , \tag{27}$$

and consider using this instead of $\Phi = 0$ as the boundary condition on $X = \pm D_x$ and $Y = \pm D_Y$ (the value of Q_∞ is of course unknown in advance but could be incorporated iteratively).

What do you conclude about the variation with L of the (non-dimensionalised) absolute error $Q_\infty - L$ and relative error $(Q_\infty - L)/L$ in (5) as an approximation to the *exact* infinite-domain plate charge per unit length? Can you account for it?

[*Hint: Question 1 may be relevant.*]

[‡] A titbit of information: according to the 80-year-old reference [1], in which R corresponds to L here and C to $Q_\infty/4\pi$, for $L = 2$ the exact analytic solution (for an infinite domain) gives $Q_\infty = 3.266$ to four figures; an up-to-date recalculation found 3.26347.

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References

1. H. B. Palmer, The Capacitance of a Parallel-Plate Capacitor by the Schwartz-Christoffel[§] Transformation, *Transactions of the American Institute of Electrical Engineers*, vol.56(3), pp.363-366 (1937). See http://dx.doi.org/10.1109/T-AIEE.1937.5057547.

2. J. C. Maxwell, *A Treatise on Electricity and Magnetism*, Part 1, Chapter XII, § 202 (1873). See https://en.wikisource.org/wiki/A_Treatise_on_Electricity_and_Magnetism.

3. S. D. Conte and C. de Boor, *Elementary Numerical Analysis*, McGraw-Hill, 1980.

4. C.-E. Fröberg, *Numerical Mathematics*, Benjamin-Cummings, 1985.

5. A. Iserles, *A First Course in the Numerical Analysis of Differential Equations*, CUP, 1996.

6. W. H. Press *et al.*, *Numerical Recipes: The Art of Scientific Computing*, CUP.

[§]Schwarz-Christoffel actually, after the German mathematician Karl Hermann Amandus Schwarz (1843-1921) – not to be confused with the French mathematician Laurent-Moïse Schwartz (1915-2002).

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